



When Cooperation Drives Continuation

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Repeated Prisoner's Dilemma

- **Prisoner's Dilemma:** dominant to defect, but repeated play can sustain cooperation through a system of threats
- In some settings, cooperation affects not only **current payoffs** but also **whether the relationship continues at all**
 - **arms negotiations:** cooperation yields gains and lowers the risk of catastrophic conflict
 - **media organizations:** successful fundraising builds financial resilience, reducing the probability of closure
- I develop a novel variant of the IRPD in which the probability of continuation is endogenously determined by the number of players who choose to cooperate in a given round
- **Two main questions:**
 - Do subjects **internalize** cooperation-contingent continuation probabilities?
 - Is the effect **comparable in magnitude** to a higher reward from mutual cooperation?

The Game

There are 2 players. For each player, the action set is $\{C(\text{oooperate}), D(\text{efect})\}$. The stage-game payoffs are symmetric and can be represented in a game matrix.

	C	D	
C	R, R	S, T	$T > R > P > S$
D	T, S	P, P	

For this game to be a Prisoner's Dilemma, it must be that $T > R > P > S$, and also I require $2R > T + S$. With such payoffs, the stage-game dominant action is (D)efection, while it is Pareto dominated by mutual (C)ooperation. The game continues for an additional round with probability $\delta : \{0, 1, 2\} \rightarrow [0, 1)$, where $\delta(c)$ denotes the probability of an additional round when c players cooperated.

	C	D	
C	$\delta(2)$	$\delta(1)$	$\delta(2) \geq \delta(1) \geq \delta(0)$
D	$\delta(1)$	$\delta(0)$	

Equilibria

- **Always Defect (AD)** can be supported as a Subgame Perfect Nash Equilibrium (SPE) if

$$\frac{1 - \delta(1)}{1 - \delta(0)} P \geq S$$

- AD is not always an equilibrium: a unilateral deviation to cooperation increases the continuation probability (from $\delta(0)$ to $\delta(1)$), thereby raising the expected value even with lower round payoff ($S < P$).

- **Grim** can be supported as an SPE if

$$\underbrace{\frac{R}{1 - \delta(2)}}_{\text{cooperative payoff}} \geq \underbrace{T + \delta(1) \frac{P}{1 - \delta(0)}}_{\text{deviation payoff}}$$

$\delta(2)$ might serve as a substitute for R as the attractiveness of cooperation can be affected by both parameters.

Experiment

	C	D
C	R, R	12, 50
D	50, 12	25, 25

(a) Stage-game
(Dal Bó & Fréchette, 2011)

	C	D
C	$\delta(2)$	1/2
D	1/2	1/2

(b) Probability of
additional round

	Grim	AD
Grim	$\frac{R}{1 - \delta(2)}, \frac{R}{1 - \delta(2)}$	37, 75
AD	75, 37	50, 50

(c) Expected payoffs

- Focus only on AD and Grim as empirically relevant strategies
- The game parameters and treatment conditions are deliberately chosen to **vary only the expected value of successful mutual cooperation** in a simplified strategy space (Blonski & Spagnolo, 2015; Dal Bó & Fréchette, 2011).
- I implement a 2×2 between-subject design with the following treatments:

$\delta(2) \backslash R$	32	48
1/2	Low-Fixed	High-Fixed
2/3	Low-Dependent	High-Dependent

- Sessions took place at the Interdisciplinary Experimental Laboratory (IELab) at Indiana University Bloomington
- Each treatment consisted of 5 sessions with 8–12 participants, totaling 224 participants
- Subjects were randomly rematched with a new partner at the end of each repeated game
- Sessions continued until the first repeated game concluded after 30 minutes of play, thereby determining the end of the session
- Participants were paid based on their cumulative earnings from all rounds of one randomly selected repeated game
 - Participants received a \$5 show-up fee and earned, on average, an additional \$13.97 based on their experimental choices

Hypotheses

- If cooperation can be supported, then I apply the concept of robustness to strategic uncertainty as an equilibrium-selection criterion (Dal Bó & Fréchette, 2018)
- **The higher the cooperation value, the more likely subjects cooperate**
- The Table below shows the inequalities for hypothesized cooperation rates

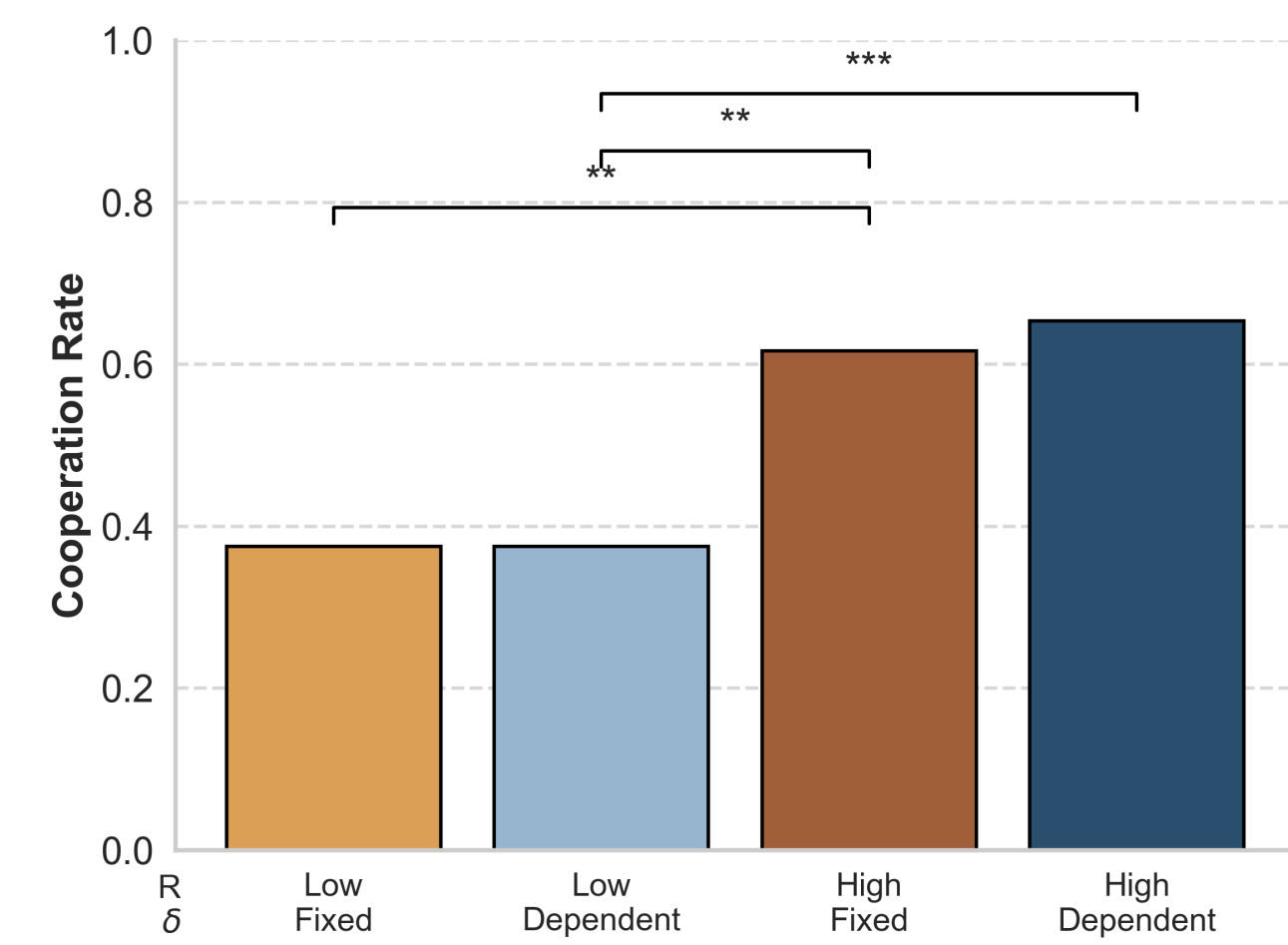
$\delta(2) \backslash R$	32		48
1/2	Low-Fixed	<	High-Fixed
	$\wedge$	$\neq$	$\wedge$
2/3	Low-Dependent	<	High-Dependent

- Cooperation higher when cooperative play is supportable as SPE
- Cooperation increases with $\delta(2)$ or R when Grim is risk-dominant
- **No change when $\delta(2) \downarrow$ and $R \uparrow$ keep cooperative payoff constant**

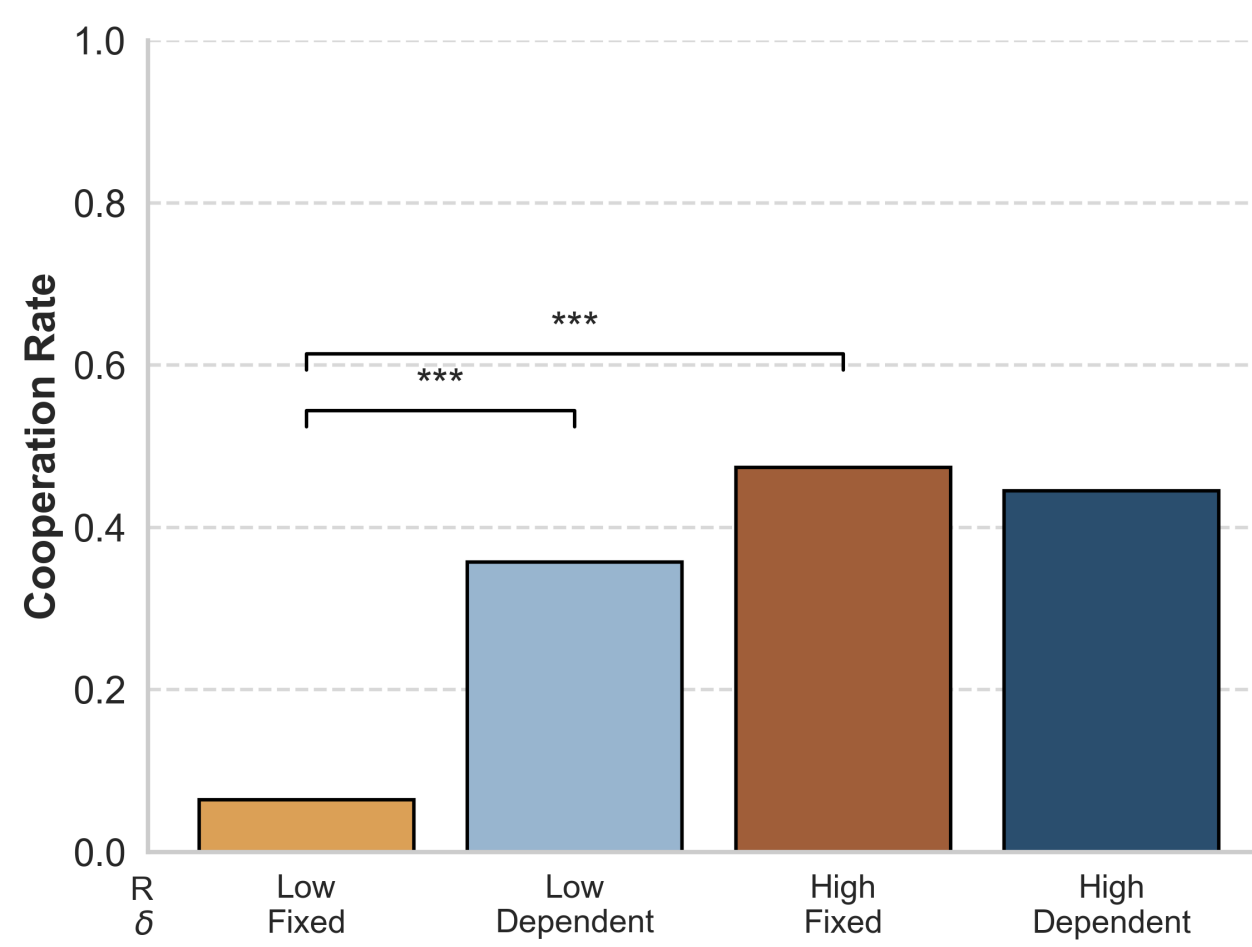
Results

Analyzing first-round behavior is essential because, under random termination, supergames vary in length across treatments, and cooperation may evolve over the course of a supergame. Main patterns:

1. A higher continuation probability raises cooperation when $R = 32$, but not when $R = 48$
2. A higher cooperative reward raises cooperation under both continuation rules
3. Holding expected cooperative payoffs constant, a higher reward yields a larger point estimate than contingent continuation



(a) First repeated game



(b) All repeated games

Figure: First-round cooperation rates

In the first round of the first supergame, increasing the reward from mutual cooperation (R) raised cooperation, whereas increasing the continuation probability following mutual cooperation did not lead to more cooperation.

Summary

- This paper studies how cooperation-contingent continuation probabilities affect cooperation in the indefinitely repeated prisoner's dilemma, using a design that independently varies the reward from mutual cooperation and the continuation probability following it
- Early in the experiment, subjects respond more strongly to increases in the reward from mutual cooperation than to increases in the cooperation-contingent continuation probability
- This asymmetry fades with experience

Key takeaways

- The endogenous-horizon game developed here offers a flexible platform for studying intertemporal incentives in repeated interactions
- The findings suggest potential implications for applied contexts. For instance, repeated public good fundraising efforts might be more effective **from the strategic point of view** when they emphasize the immediate reward of contributing, rather than appealing to the increased likelihood of survival following a donation.